

Calculation of Radial Diffusion Coefficients using combined data from RBSP and THEMIS missions

Konstantina Thanasoula¹, Christos Katsavrias¹, Ioannis A. Daglis¹, Constantinos Papadimitriou², Aaron W. Breneman³, Sheng Tian⁴, Georgios Balasis⁵ and Theodore Sarris⁶

1. Department of Physics, National and Kapodistrian University of Athens, Greece
2. SPARC – Space Applications and Research Consultancy, Athens, Greece
3. Goddard Space Flight Center, NASA, Greenbelt, MD, USA
4. Department of Atmospheric and Oceanic Sciences, University of California, Los Angeles, CA, USA
5. IAASARS, National Observatory of Athens, Greece
6. Department of Electrical and Computer Engineering, Democritus University of Thrace, Greece

Abstract

Radial diffusion plays a key role in the dynamics of the outer Van Allen belt. In this study, we estimate radial diffusion coefficients (D_{LL}) using the Fei et al. (2006) formalism. Our goal is to create a comprehensive database that includes Power Spectral Density (PSD) and D_{LL} data from multiple missions, thereby increasing both spatial and temporal coverage. In this way, the uncertainty caused by using only single-mission data, which could lead to under- or overestimation of radial diffusion coefficients, is reduced. We initially use magnetic and electric field data from the THEMIS and RBSP missions, and our future goal is to include additional missions such as Arase, in order to further extend the coverage of the database. The derived products will be added to the existing “SafeSpace” database, which includes PSD and D_{LL} computed from THEMIS mission data over a nine-year period (<https://synergasia.uoa.gr/modules/document/?course=PHYS120>). Our results indicate that when spacecraft from the two missions, RBSP and THEMIS, are located at the same L^* , there is a strong correlation between their magnetic field measurements. Using the combined datasets, we estimate proxies for both the electric and magnetic components of D_{LL} . The inclusion of RBSP data provides a better representation of the D_{LL} proxies at $L^* < 6$. The combined dataset results in more uniform D_{LL} proxy distributions in L^* and MLT, particularly under disturbed geomagnetic conditions, reducing the uncertainties associated with limited spatial coverage. Overall, the increased coverage in MLT and L^* provides a more comprehensive calculation of ULF wave PSD and consequently, a more accurate estimation of D_{LL} .

Introduction

Radial diffusion plays a fundamental role in both the acceleration and loss of particles in the outer Van Allen belt. It occurs through the violation of the third adiabatic invariant due to wave-particle interactions, particularly with Ultra Low Frequency (ULF) waves in the Pc4-Pc5 range, and is described by the Fokker–Planck equation, which in its simplest form is given by:

$$\frac{\partial f}{\partial t} = L^2 \frac{\partial}{\partial L} \left(\frac{D_{LL}}{L^2} \frac{\partial f}{\partial L} \right)$$

where L is the Roederer L^* parameter, and D_{LL} is the radial diffusion coefficient, which is defined as the mean square rate of change of L^* for a large number of particles over time. One of the most widely used formalisms for the D_{LL} calculation is that of Fei et al. (2006), according to which D_{LL} consists of both an electric component, which is related to azimuthal field disturbances and is influenced by ULF waves created due to Kelvin-Helmholtz instabilities at the magnetospheric flanks, and a magnetic component, which is associated with disturbances in the parallel magnetic field and is influenced by ULF waves resulting from fluctuations in the solar wind dynamic pressure. This formalism assumes uniform power distribution in azimuth, but in-situ measurements show that the azimuthal distribution depends on the physical origin of the waves and their propagation characteristics. Single-mission measurements provide limited MLT coverage, and therefore we combine RBSP and THEMIS data to increase spatial and temporal coverage and improve D_{LL} estimation.

References

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Contact info:

konstantina.thanasoula93@gmail.com

Data & Methods

In order to create a comprehensive database, we used magnetic and electric field measurements from the RBSP A&B and THEMIS A,D and E satellites. The measurements of the two missions were first compared as a function of L^* , Magnetic Local Time (MLT), B_{ratio} ($B_{equatorial}/B_{local}$), and geomagnetic activity (Kp index) to assess their consistency, focusing only on near-equatorial measurements ($B_{ratio}>0.8$). Ephemeris parameters were calculated using the Olson-Pfitzer Quiet (1977) model. Subsequently, the measurements were transformed into Mean Field Aligned (MFA) coordinates, in which the mean field direction was defined using a 30-min running average of the magnetic field vector.

Magnetic Field:

The RBSP and THEMIS magnetic field measurements show a strong correlation at the same L^* values. Following the MFA transformation, the PSD was calculated as a function of time and frequency, while the total power was calculated as a function of time (Figure 1). Spatial gradients near perigee, caused by spacecraft motion, were identified and removed from the datasets.

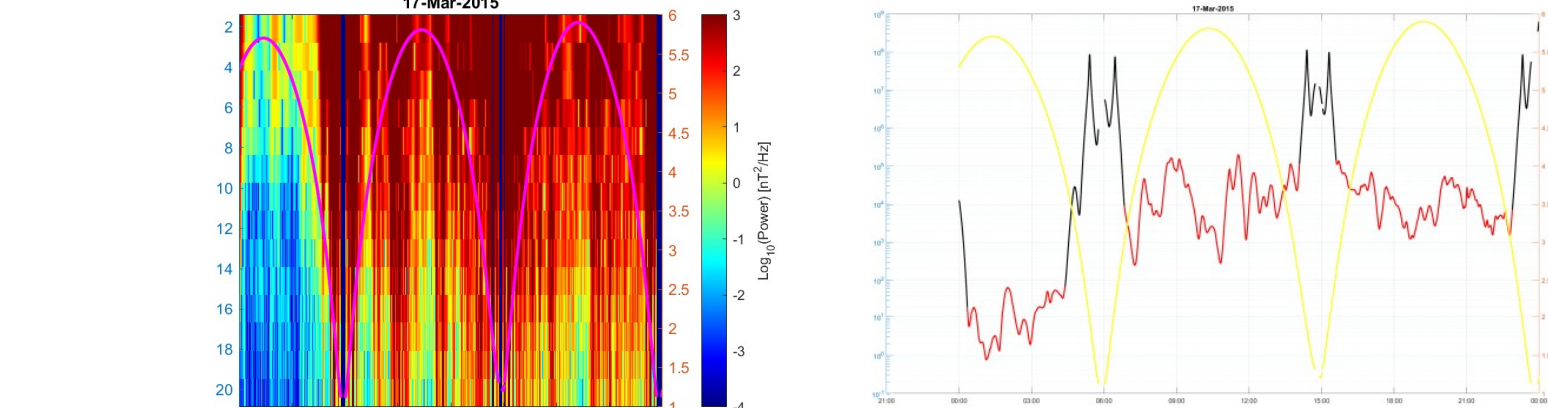


Figure 1: RBSP-A measurements during the St. Patrick's event (March 17, 2015). Left panel: PSD as a function of time and frequency (left axis). L^* is shown on the right axis, and the satellite's orbit is shown in pink. Right panel: total power as a function of time. L^* is shown on the right axis, and the satellite's orbit is shown in yellow.

Electric Field:

The comparison between RBSP and THEMIS electric field measurements indicates significant discrepancies, which were reduced by multiplying the RBSP toroidal component by a factor of 4 (Figure 2).

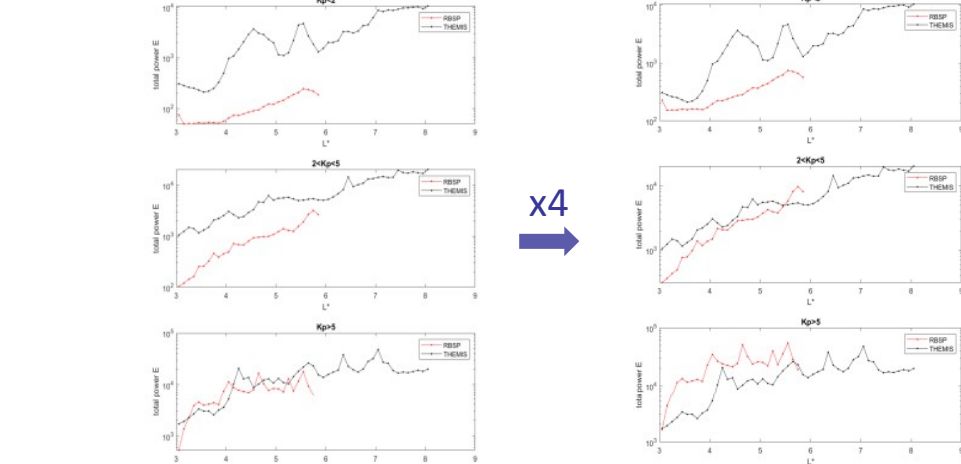


Figure 2: Total wave power of the toroidal component for different levels of geomagnetic activity: quiet ($Kp \leq 2$), moderate ($2 < Kp \leq 5$) and disturbed ($Kp > 5$). Left panel: original total power. Right panel: total power after multiplying the RBSP toroidal component by a factor of 4.

Subsequently, the electric and magnetic radial diffusion coefficient proxies (D_{LL}^E and D_{LL}^B , respectively) were calculated. The D_{LL} proxy was calculated as the product of the weighted averaged power with a simple multiplication factor.

Results

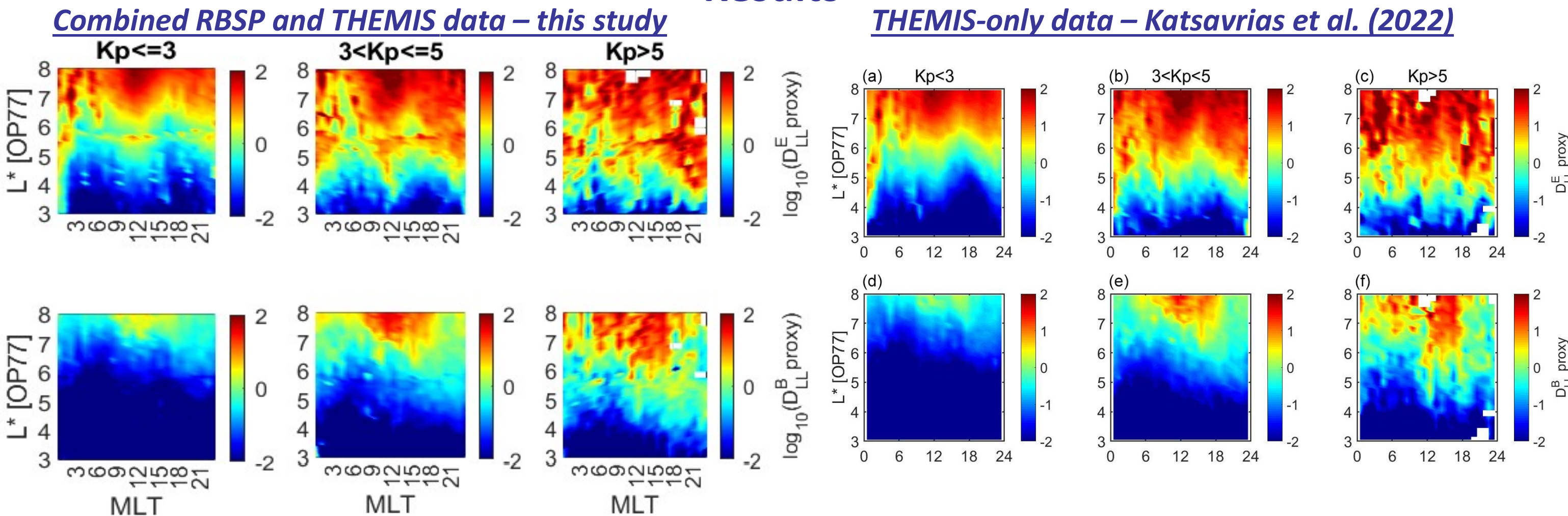


Figure 3: D_{LL}^E (top) and D_{LL}^B (bottom) proxies as a function of L^* and MLT for quiet (left), moderate (middle), and disturbed (right) geomagnetic conditions.

Figure 4: THEMIS-only D_{LL} proxy distributions from Katsavrias et al. (2022), shown for comparison.

Conclusions

- Compared with the THEMIS-only results of Katsavrias et al. (2022), the inclusion of RBSP data improves the representation of D_{LL}^E and D_{LL}^B proxies at $L^* < 6$.
- The inclusion of RBSP data increases the MLT coverage per hour and L^* bin, reaching a maximum coverage of 6 hours using the five satellites.
- Combining RBSP and THEMIS data provides more uniform D_{LL} proxy distributions in MLT, especially under disturbed geomagnetic conditions, reducing uncertainties associated with limited MLT coverage.

Ongoing & Future Work

- Extend the spatial and temporal coverage of the “SafeSpace” database by including additional missions, such as Arase.
- Repeat the analysis using the TS04 magnetic field model (Tsyganenko and Sitnov, 2005).